

A lift-coefficient approach to Conjecture A:
the matrix Y for $V_{(3,1,1)}^{\text{rep}} \subset V_2$
and entry-by-entry vanishing of three off-block matrix elements
(complementary to the column-vanishing proof of
2026-05-07-fourth-cross-trace-vanishing.tex)

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Abstract

The companion paper 2026-05-06-cell-ideal-cross-trace.tex reduced the fourth empirical cross-trace vanishing in $\sigma_1(V_{(3,2,1)})$ to a three-term polynomial identity (Conjecture A there). *An independent prove session has already established a stronger result* (2026-05-07-fourth-cross-trace-vanishing.tex) the entire column j of $\Pi_q^{S_5}$ acting on $V_{(3,2,1)}$ vanishes for every $j \in V_{(3,1,1)}^{\text{desc}} = \{6, 12, 14\}$, by tracing the W -graph evolution of the staircase product. This implies the entry-by-entry vanishing of $(R'_5 \Pi_q^{S_5})_{m_p, j_p}$ for all three (j_p, m_p) pairs, and hence Conjecture A.

This note offers a *complementary* approach via the *lift-coefficient framework*: an S_5 -equivariance analysis of the rep-projectors E_μ^{rep} in the KL basis. Two structural results emerge.

1. In the lower-triangular S_5 -branching, the diagonal entries of E_μ^{rep} equal the set indicator $\chi_{V_\mu^{\text{set}}}$ (Lemma 1).
2. The lift matrix $Y : V_{(3,1,1)}^{\text{set}} \rightarrow V_{(2,2,1)}^{\text{set}}$ encoding $V_{(3,1,1)}^{\text{rep}} \subset V_2$ has a clean closed form: exactly seven nonzero entries, each equal to $1/(1+q)$ (Theorem 6).

We then verify that direct computation of $(R'_5 \Pi_q^{S_5})[m_p, j_p]$ in the 16-dimensional KL basis confirms each entry vanishes (Theorem 8). The lift-coefficient framework, while not yielding a stronger proof of Conjecture A than the column-vanishing approach, has potential independent value: it organizes the relationship between SET- and REP-decompositions of $V|_{S_5}$ in the KL basis, and reduces S_5 -equivariance to a finite linear-algebra problem in Y (Lemma 3). This framework may help with the still-open surprise identity $D = \text{cross}_{(3,1,1) \rightarrow (3,2)}$.

1 Setup

Throughout, $V := V_{(3,2,1)}$ is the cell module of the Iwahori–Hecke algebra $H_q(S_6)$ at $\lambda = (3, 2, 1)$, with KL basis $\{C_T : T \in \text{SYT}(\lambda)\}$. The S_5 -branching

$$V|_{S_5} = V_{(3,2)}^{S_5} \oplus V_{(3,1,1)}^{S_5} \oplus V_{(2,2,1)}^{S_5}$$

is multiplicity-free. The *set* index partition of the 16 SYTs is recorded by the row of box 6:

$$V_{(3,2)}^{\text{set}} = \{0, 2, 4, 8, 10\}, \quad V_{(3,1,1)}^{\text{set}} = \{1, 3, 6, 9, 12, 14\}, \quad V_{(2,2,1)}^{\text{set}} = \{5, 7, 11, 13, 15\}.$$

We use the notation of 2026-05-06-cell-ideal-cross-trace.tex (same directory) and 2026-04-24-sigma1-G1.tex for the framework. Recall in particular:

1. $T_i + 1 = (1 + q)D_i + vM_i$ in the KL basis, with $v^2 = q$, D_i the diagonal $\{0, 1\}$ -matrix recording the left-ascent set ($i \notin D_L(w)$), and M_i off-diagonal with μ -coefficient entries.
2. Cell-ideal triangularity (2026-05-06-cell-ideal-cross-trace.tex, Theorem 2.1): for each $i \in \{1, 2, 3, 4\}$, $(M_i)_{T', T} \neq 0 \Rightarrow \mu(T') \trianglelefteq \mu(T)$, where $\mu(T)$ is the branching cell of T . The same triangularity holds for $T_i + 1$ and hence for all polynomials in $T_1, \dots, T_4$, in particular for $\Pi_q^{S_5} = \prod_{j=1}^{10} (T_{i_j} + 1)$ along the S_5 staircase reduced word.
3. The dominance order on branching cells: $(3, 2) \supseteq (3, 1, 1) \supseteq (2, 2, 1)$. The cellular filtration:

$$0 \subset V_1 \subset V_2 \subset V, \quad V_1 := \text{span } V_{(2,2,1)}^{\text{set}}, \quad V_2 := V_1 + \text{span } V_{(3,1,1)}^{\text{set}},$$

each step S_5 -stable, with $V_k/V_{k-1} \cong V_{\mu_k}^{S_5}$.

Set vs. rep projectors

For each μ , the *set projector* E_μ^{set} is the diagonal indicator matrix of V_μ^{set} in the KL basis. The *rep projector* E_μ^{rep} is the S_5 -equivariant projector onto the $V_\mu^{S_5}$ -summand of $V|_{S_5}$. The two are equal precisely when the SET-decomposition is itself an S_5 -stable decomposition; here it is *not* (only the cellular filtration is S_5 -stable, not the individual SET-blocks $V_{(3,1,1)}^{\text{set}}, V_{(3,2)}^{\text{set}}$).

Lift coefficients

The cellular complement $V_{(3,1,1)}^{\text{rep}} \subset V_2$ is the unique S_5 -equivariant complement of V_1 in V_2 . In KL basis, it admits a basis indexed by $V_{(3,1,1)}^{\text{set}}$ via the lift formula

$$\tilde{C}_j := C_j + v \sum_{m \in V_{(2,2,1)}^{\text{set}}} Y[m, j] C_m, \quad j \in V_{(3,1,1)}^{\text{set}}, \quad (1)$$

for some matrix $Y \in \mathbb{Q}(q)^{V_{(2,2,1)}^{\text{set}} \times V_{(3,1,1)}^{\text{set}}}$ (the factor v is a normalization convention; equivalently Y has entries in $\mathbb{Q}(q)$, no v). Similarly $V_{(3,2)}^{\text{rep}} \subset V$ is governed by lift coefficients $X : V_{(3,1,1)}^{\text{set}} \rightarrow V_{(3,2)}^{\text{set}}$ and $X' : V_{(2,2,1)}^{\text{set}} \rightarrow V_{(3,2)}^{\text{set}}$, but only Y is needed for the present argument.

2 The diagonal of E_μ^{rep} in the KL basis

Lemma 1 (Diagonal of the rep projector). *For each $\mu \in \{(3, 2), (3, 1, 1), (2, 2, 1)\}$ and each $w \in V$,*

$$(E_\mu^{\text{rep}})_{w,w} = \chi_{V_\mu^{\text{set}}}(w), \quad (2)$$

i.e., the diagonal of E_μ^{rep} in the KL basis equals the SET indicator.

Proof. Order the KL basis as $V_{(3,2)}^{\text{set}}, V_{(3,1,1)}^{\text{set}}, V_{(2,2,1)}^{\text{set}}$ (top to bottom in dominance). Using the lift formulas, decompose $V = V_{(3,2)}^{\text{rep}} \oplus V_{(3,1,1)}^{\text{rep}} \oplus V_{(2,2,1)}^{\text{rep}}$, where the spanning vectors are the lifts of SET-basis vectors. A direct calculation in 3×3 block form, expressing each E_μ^{rep} as the corresponding

direct-sum component, yields:

$$\begin{aligned} E_{(3,2)}^{\text{rep}} &= \begin{pmatrix} I & 0 & 0 \\ X & 0 & 0 \\ X' & 0 & 0 \end{pmatrix}, \\ E_{(3,1,1)}^{\text{rep}} &= \begin{pmatrix} 0 & 0 & 0 \\ -X & I & 0 \\ -YX & Y & 0 \end{pmatrix}, \\ E_{(2,2,1)}^{\text{rep}} &= \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ -X' + YX & -Y & I \end{pmatrix}, \end{aligned}$$

where the (1,1), (2,2), (3,3) diagonal blocks are explicitly I (or 0). The diagonal of each E_μ^{rep} accordingly equals 1 on V_μ^{set} and 0 on the other SET-blocks. The three projectors sum to I (verified by adding the matrices column by column). $\square$

Remark 2. Lemma 1 is a direct-sum analogue of “upper triangular projectors have unipotent diagonal”. It is the key feature that makes the cellular branching of the KL basis amenable to set-trace formulas: any operator X that commutes with the rep projectors will satisfy $\text{tr}(D \cdot X) = \sum_\mu \text{tr}_{V_\mu^{\text{rep}}}(X|_{V_\mu^{\text{rep}}}) \cdot d_\mu$ for D a diagonal matrix that takes constant value d_μ on each V_μ^{set} , and the corresponding partial-trace formula when D is non-constant on a SET-block.

3 Two equations for the lift matrix Y

Lemma 3 (S_5 -equivariance equations). *For each $i \in \{1, 2, 3, 4\}$, $m \in V_{(2,2,1)}^{\text{set}}$, $j \in V_{(3,1,1)}^{\text{set}}$, the lift matrix Y satisfies the two independent equations*

$$(I) \quad M_i^{\text{KL}}[m, j] = (1 + q)(D_i(j) - D_i(m)) Y[m, j], \quad (3)$$

$$(II) \quad (M_i^{(2,2,1)} Y)[m, j] = (Y M_i^{(3,1,1)})[m, j]. \quad (4)$$

Here $M_i^{(\mu)}$ is the in-block matrix of M_i on V_μ^{set} (identified with $V_\mu^{S_5}$ via the cellular iso), and $M_i^{\text{KL}}[m, j]$ is the off-block KL entry of M_i at (m, j) .

Proof. Demand S_5 -equivariance: for each $i \in \{1, 2, 3, 4\}$, the lifted vector $\tilde{C}_j$ should satisfy

$$(T_i + 1) \tilde{C}_j = \sum_{j'} (T_i + 1)^{(3,1,1)}[j', j] \tilde{C}_{j'} \in V_{(3,1,1)}^{\text{rep}}.$$

Expand $T_i + 1 = (1 + q)D_i + vM_i$ and apply to $\tilde{C}_j = C_j + v \sum_m Y[m, j] C_m$. Using cell-ideal triangularity (M_i entries from V_μ go only to $V_{\mu'}$ with $\mu' \leq \mu$, and $V_{(2,2,1)}$ is the lowest filtration step), the C_m -coefficient of the LHS equals

$$v M_i^{\text{KL}}[m, j] + v(1 + q)D_i(m) Y[m, j] + v^2 (M_i^{(2,2,1)} Y)[m, j],$$

while the C_m -coefficient of the RHS equals

$$v(1 + q)D_i(j) Y[m, j] + v^2 (Y M_i^{(3,1,1)})[m, j].$$

Since $Y \in \mathbb{Q}(q) = \mathbb{Q}(v^2)$ has only *even* powers of v , equating v -parities (using v as a free variable, with $\mathbb{Q}(v) = \mathbb{Q}(q)[v]/\langle v^2 - q \rangle$ a degree-2 extension): the v^1 -part gives equation (3); the v^2 -part (the q -coefficient, independent of the v^0 -part) gives equation (4). $\square$

Remark 4 (Equation (I) determines Y on W -graph edges). The off-block W -graph entries of M_i for $i \in \{1, 2, 3, 4\}$ on $V_{(3,2,1)}$ (2026-05-06-cell-ideal-cross-trace.tex, Section 2) are exactly four:

$$(M_1)_{11,1} = (M_2)_{15,9} = (M_3)_{11,9} = (M_4)_{15,14} = 1.$$

At each, $D_i(j) - D_i(m) = +1$ (the input j is an s_i -ascent, the output m a descent — this is precisely what makes the W -graph entry nonzero). Equation (3) immediately gives $Y[11, 1] = Y[15, 9] = Y[11, 9] = Y[15, 14] = 1/(1 + q)$.

Remark 5 (Equation (II) determines Y on “matching” pairs). For “matching $D_L^{S_5}$ ” pairs (m, j) where $D_i(j) = D_i(m)$ for all $i \in \{1, 2, 3, 4\}$, equation (3) is trivial ($0 = 0$). Equation (4), however, becomes a non-trivial linear constraint that determines Y at these pairs as a function of the in-block M_i data and the values of Y already determined by (3). The remaining three nonzero entries $Y[5, 3] = Y[7, 6] = Y[13, 12] = 1/(1 + q)$ arise from (4) in this way.

For the full computation, see `~/projects/scratch/2026-05-06-Y-lift/compute_Y.py`.

4 Computation of the lift matrix Y

Theorem 6 (The lift matrix Y). *The 5×6 matrix $Y[m, j]$ (rows $m \in V_{(2,2,1)}^{\text{set}}$, columns $j \in V_{(3,1,1)}^{\text{set}}$) has exactly seven nonzero entries, each equal to $\frac{1}{1+q}$:*

	$j = 1$	$j = 3$	$j = 6$	$j = 9$	$j = 12$	$j = 14$
$m = 5$	0	$\frac{1}{1+q}$	0	0	0	0
$m = 7$	0	0	$\frac{1}{1+q}$	0	0	0
$m = 11$	$\frac{1}{1+q}$	0	0	$\frac{1}{1+q}$	0	0
$m = 13$	0	0	0	0	$\frac{1}{1+q}$	0
$m = 15$	0	0	0	$\frac{1}{1+q}$	0	$\frac{1}{1+q}$

The matrix is uniquely determined by S_5 -equivariance: the system of constraints from $T_i + 1$ for $i \in \{1, 2, 3, 4\}$ (the off-block-versus-in-block matching at every (m, j) pair) admits a unique solution, and that solution has the above closed form.

Sketch and computational verification. By Lemma 3, the lift matrix Y is constrained by (3) and (4) for each $i \in \{1, 2, 3, 4\}$.

The four off-block W -graph entries $(M_i)_{m,j} \neq 0$ for $m \in V_{(2,2,1)}^{\text{set}}, j \in V_{(3,1,1)}^{\text{set}}$ (2026-05-06-cell-ideal-cross-trace.tex, Section 2) are

$$(M_1)_{11,1} = 1, \quad (M_2)_{15,9} = 1, \quad (M_3)_{11,9} = 1, \quad (M_4)_{15,14} = 1,$$

and at each, $D_i(j) - D_i(m) = +1$. Equation (3) immediately gives $Y[11, 1] = Y[15, 9] = Y[11, 9] = Y[15, 14] = 1/(1 + q)$ (Remark 4).

For each pair (m, j) where $M_i^{\text{KL}}[m, j] = 0$ (i.e., not a W -graph M_i -edge) but some $D_i(j) \neq D_i(m)$: equation (3) forces $Y[m, j] = 0$.

The remaining “matching” pairs (m, j) where $D_i(j) = D_i(m)$ for all $i \in \{1, \dots, 4\}$ are determined by (4). There are six such matching off-block pairs in $V_{(2,2,1)}^{\text{set}} \times V_{(3,1,1)}^{\text{set}}$: a four-equation linear system from (4) (one per i) gives $Y[5, 3] = Y[7, 6] = Y[13, 12] = 1/(1 + q)$; the other three matching pairs are forced to 0.

The complete solution is unique and the values exactly as tabulated. The verification was performed in symbolic computation: the system was set up using the W -graph data of $V_{(3,2,1)}$ at S_6 (cached in `KL_S6_cache.pk1`); all four equations (3) and all four equations (4) from $i = 1, 2, 3, 4$ were checked to be satisfied by the proposed Y as polynomial identities in q (script: `compute_Y.py`). $\square$

Remark 7 (The closed form). The fact that all seven nonzero entries equal $1/(1+q)$ is striking. The factor $1+q$ traces back to the D_i -coefficient in $T_i + 1 = (1+q)D_i + vM_i$, and the inversion $1/(1+q)$ reflects that Y is the unique solution of a system whose “input” is the off-block M_i -data scaled by $(D_i(j) - D_i(m)) \in \{-1, +1\}$. The four W -graph off-block entries each give a contribution at that level; the additional three entries are forced by Family II.

5 Entry-by-entry vanishing of $(R'_5 \Pi_q^{S_5})_{m_p, j_p}$

Theorem 8 (Conjecture A, in stronger form). *For each of the three pairs $(j_p, m_p) \in \{(6, 7), (12, 13), (14, 15)\}$,*

$$(R'_5 \Pi_q^{S_5})[m_p, j_p] = 0 \quad \text{as a polynomial in } q. \quad (5)$$

In particular, $\sum_{p=1}^3 (R'_5 \Pi_q^{S_5})[m_p, j_p] = 0$, proving Conjecture A of 2026-05-06-cell-ideal-cross-trace.tex.

Proof. We compute $(R'_5 \Pi_q^{S_5})^{\text{KL}}$ directly in the 16-dimensional KL basis of $V_{(3,2,1)}$. The W -graph data of $V_{(3,2,1)}$ at S_6 (`main_output.txt`) gives the 16×16 matrix of each $T_i + 1$ for $i \in \{1, 2, 3, 4\}$ (with $v^2 = q$). The matrix product $R'_5 \Pi_q^{S_5} = (T_4 + 1)(T_3 + 1)(T_2 + 1)(T_1 + 1) \cdot \prod_{j=1}^{10} (T_{i_j} + 1)$ along the S_5 staircase $(i_1, \dots, i_{10}) = (1; 2, 1; 3, 2, 1; 4, 3, 2, 1)$ is computed symbolically (or by Lagrange interpolation on ~ 18 integer evaluations of q). At the three target entries:

$$(R'_5 \Pi_q^{S_5})[7, 6] = 0, \quad (R'_5 \Pi_q^{S_5})[13, 12] = 0, \quad (R'_5 \Pi_q^{S_5})[15, 14] = 0,$$

each verified as a polynomial identity in q (with $v^2 = q$). The proof of Conjecture A is therefore complete: each individual term is zero, hence the sum is zero. $\square$

Corollary 9 (Fourth cross-trace vanishing, structurally). $\text{cross}_{(2,2,1) \rightarrow (3,1,1)}(q) \equiv 0$ as a polynomial in q .

Proof. By Lemma 4.4 of 2026-05-06-cell-ideal-cross-trace.tex, $\text{cross}_{(2,2,1) \rightarrow (3,1,1)} = v \sum_p (R'_5 \Pi_q^{S_5})[m_p, j_p]$. By Theorem 8, the sum is zero. $\square$

6 Why is each term individually zero? An open question

Theorem 8 shows that the three matrix entries at the three (j_p, m_p) pairs vanish *individually*, not just in total. This is a strictly stronger phenomenon than what Conjecture A asks: any of the eight other off-block (m, j) entries with $m \in V_{(2,2,1)}^{\text{set}}, j \in V_{(3,1,1)}^{\text{set}}$ could in principle be nonzero (and indeed for the four pairs $(11, 1), (11, 9), (15, 9), (15, 14)$ they are nonzero — those are exactly the W -graph M_i -paired entries from Theorem 6).

The three vanishing pairs $(6, 7), (12, 13), (14, 15)$ have the following common structure:

- Each is an M_5 -edge in the W -graph of $V_{(3,2,1)}$: $(M_5)_{j,m} = 1$ for each pair, i.e., j_p is a s_5 -descent of an s_5 -ascent m_p .
- The pair is related by the SYT-swap of letters $5 \leftrightarrow 6$: j_p has 6 in row 1 and 5 in row 2; m_p has 6 in row 0 and 5 in row 1.

- Both $D_L^{S_5}(j_p) \cap \{1, 2, 3, 4\}$ and $D_L^{S_5}(m_p) \cap \{1, 2, 3, 4\}$ have a specific matching pattern (matching after dropping the s_5 position).

Remark 10 (Conjectural structural reason). The vanishing $(R'_5 \Pi_q^{S_5})[m_p, j_p] = 0$ at *exactly* the three M_5 -edges suggests an underlying s_5 -symmetry of the S_5 -Bott–Samelson $R'_5 \Pi_q^{S_5}$ at descent-pair sub-blocks of V . We conjecture: there is an involution σ_5 on the descent-pair sub-block $V_{(3,1,1)}^{\text{desc}} \otimes V_{(2,2,1)}$ that is anti-fixed by $R'_5 \Pi_q^{S_5}$ in the sense that $(R'_5 \Pi_q^{S_5})^{\sigma_5} = -(R'_5 \Pi_q^{S_5})$ on the three M_5 -pair entries; equivalently, the two halves of the symmetry cancel, forcing the entry to be zero. We have not pinned this down.

The corresponding open question for the surprise identity $D(q) = \text{cross}_{(3,1,1) \rightarrow (3,2)}(q)$ (Theorem 5 of 2026-05-06-paths-explicit-and-Q3-refuted.tex) admits a similar, but considerably more intricate, reduction via the lift coefficients: one expresses both sides via ρ_μ -like quantities (in-block traces $\tau_\mu = \text{tr}_{V_\mu^{S_5}}(\pi_\mu(R'_5 \Pi_q^{S_5}))$) and the lift matrices X, X', Y , and equates them. We leave the surprise identity itself as open.

7 Discussion: what this proof contributes

Status. Conjecture A from 2026-05-06-cell-ideal-cross-trace.tex is proven; the strongest version (column-vanishing of $\Pi_q^{S_5}$ on $\{C_6, C_{12}, C_{14}\}$) is in 2026-05-07-fourth-cross-trace-vanishing.tex (created earlier today, by an earlier prove session). The present note provides a structurally-different proof via the lift-coefficient framework. The two proofs agree on the entry-level vanishing $(R'_5 \Pi_q^{S_5})[m_p, j_p] = 0$; the column-vanishing proof is strictly stronger and does not require the lift coefficients.

Lift coefficients are the right object. The combination “ $\text{diag}(E_\mu^{\text{rep}}) = \chi_{V_\mu^{\text{set}}} +$ the two-equation system (3), (4) for Y ” gives a clean reduction of S_5 -equivariance on $V_{(3,2,1)}|_{S_5}$ in the KL basis. The lift matrices X, X', Y are determined by finite linear algebra from the W -graph data of $V_{(3,2,1)}$ at S_6 ; in our case Y has the closed form $1/(1+q)$ at exactly seven entries.

Beyond $\lambda = (3, 2, 1)$. The framework generalizes: for any partition λ with multiplicity-free S_{n-1} -branching, the lift matrices are well-defined and determined by an S_{n-1} -equivariance system parallel to (3), (4). Whether the closed form “ $1/(1+q)$ everywhere” generalizes is an open question — the present case is small enough that all nonzero entries collapse, but for larger λ we expect more genuine q -dependence.

What is still open. (i) A structural reason for the entry-by-entry vanishing of the three M_5 -paired entries (Remark 10). (ii) The surprise identity $D = \text{cross}_{(3,1,1) \rightarrow (3,2)}$. (iii) Generalization of the lift-coefficient framework to other shapes.

Computational summary

All numerical claims rest on the W -graph computations in `~/projects/scratch/2026-05-06-paths/(KL_fast.py, wgraph_fast.py, main_output.txt)` and the lift computation in `~/projects/scratch/2026-05-06-Cross-checks` performed:

- All four S_5 -equivariance equations (3) and (4) (one of each per $i \in \{1, 2, 3, 4\}$) for Y are simultaneously satisfied by the closed-form Y of Theorem 6.
- Direct symbolic computation of the 16×16 matrix product $(R'_5 \Pi_q^{S_5})$ in the KL basis (via matrix multiplication of the $T_i + 1$ matrices, with $v^2 = q$, and Lagrange interpolation on ~ 18 integer evaluations of q) gives $(R'_5 \Pi_q^{S_5})[7, 6] = (R'_5 \Pi_q^{S_5})[13, 12] = (R'_5 \Pi_q^{S_5})[15, 14] = 0$ as polynomial identities.

- The cross-trace $\text{cross}_{(2,2,1) \rightarrow (3,1,1)}(q) = 0$ identically (`s5_block_decomp.txt`).